

“...THERE IS A **NEW VISIBLE WORLD DISCOVERED** TO THE UNDERSTANDING.”

Robert Hooke, English inventor, from *Micrographia*, 1665



This image of a louse is from *Micrographia*—a record of observations Robert Hooke made using a microscope.

ENGLISH INVENTOR ROBERT

HOOKE, Curator of Experiments at the Royal Society, London, had turned his attention to **microscopy**. In 1665, he published the Society's first monograph, *Micrographia*, with exquisite illustrations of miniature life, including the **first depiction of a microorganism**—in this case, a mould. The book contained the first published reference to a **biological cell**, which Hooke named after looking at cork tissue.

In this year, Cambridge University, England, was closed as a precaution against the

Ice cap on Mars

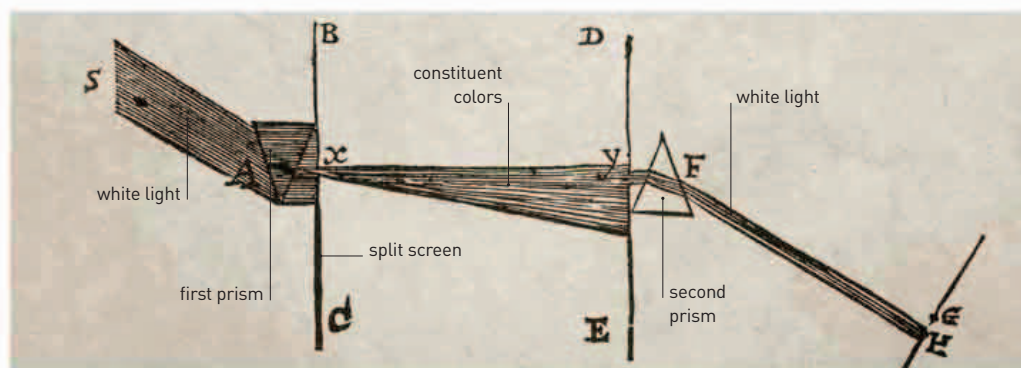
Giovanni Cassini observed an ice cap on Mars, although it was centuries later that images such as this helped reveal its makeup.

plague. One of its students, physicist and mathematician **Isaac Newton** (1642–1727), used his freedom to make extraordinary discoveries. Within two years, he invented **calculus**, and used **prisms** to study the colors of a rainbow.

In 1666, Italian astronomer **Giovanni Cassini** (1625–1712) was the first to observe that the planet Mars had a polar cap.



He also calculated the planet's rotational period to be about 24 hours, 40 minutes. In the previous two years, he had determined the rotational periods of the planets Jupiter and Venus as well.



NEWTONIAN PRISM EXPERIMENT

Although other scientists had shone white light through a prism to produce a rainbow of colors, Isaac Newton had the novel idea that the colors were constituents of white light, which were separated by the prism. He proved this by placing a second prism upside-down in front of the first.

The first prism splits white light into seven colors, each made up of light with a different wavelength. Splitting happens because light with the longest wavelength (red) bends less than light with the shortest wavelength (violet). The second prism bends them again—and so recombines them.

11 PINTS
THE AVERAGE
VOLUME OF BLOOD
IN THE ADULT
HUMAN BODY

IN 1666, THE FIRST BLOOD

TRANSFUSIONS—dog to dog—had been demonstrated before the Royal Society. In 1667, animal-to-human “therapeutic” blood transfusions were attempted; animal blood was regarded as less likely “to be rendered impure by passion or vice.” Independently of one another, English physician **Richard Lower** (1631–91) and French physician **Jean-Baptiste Denis** (1643–1704) transfused small quantities of **lamb's blood into their patients**. Those patients lucky enough to survive doubtless did so because the allergic reaction was minimal.

horizontal
eyepiece

Although the book *Experiments on the Generation of Insects* appeared as an obscure publication in 1668, its author, Italian physician **Francesco Redi** (1626–97), had described potentially groundbreaking experiments in it. Redi was testing the idea that life—specifically maggots—could form spontaneously, as was the prevailing wisdom. He placed

1 1/4 INCHES
THE DIAMETER
OF THE **OBJECTIVE MIRROR** IN **NEWTON'S TELESCOPE**

However, the procedure was eventually banned in France following a number of fatalities.

In 1668, **Isaac Newton** built the **first reflecting telescope**. By using mirrors, the design avoided the lens aberration associated with refracting telescopes. Scottish astronomer James Gregory had described such an instrument five years earlier, but had no means of producing it.

pieces of meat in jars, sealing some with gauze and leaving others open. Maggots appeared only in the open jars—evidence they could not form on their own. However, his debunking of **spontaneous generation** had little impact on the progress of biological thought

January 5, 1665
Journal des Sçavans,
the first scientific
journal, begins
publication in Paris

January 1665
English philosopher
Robert Hooke
publishes *Micrographia*

March 21, 1666 Hooke
describes gravity as
an attractive force

1666 Newton observes
light refraction by using
a prism, and invents
calculus

1666 Italian astronomer
Giovanni Cassini discovers
Mars has a polar ice cap

June 15, 1667
French physician
Jean-Baptiste Denis
performs the first
recorded human
blood transfusion

October 1667 Hooke
shows heart function
depends on inflation
of the lungs

1667 Danish
biologist Nicolas
Steno produces
the earliest
geological treatise